

Machine Learning

Homework 4

Dec. 2025

Fill your Name
Fill your Student ID

1. Here is a sample table. Please use the data in the table 1 to solve the decision tree-related problems.

Day	Outlook	Temperature	Humidity	wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Hot	High	Strong	No

Table 1: Instances for building a decision tree.

- (a) Compute the Overall Entropy.
(b) Compute the information gain and select the root node of the decision tree.

Ans:

(a)

(b)

2. Here is a sample table. Please use the data in the table 2 to solve the naive Bayes-related problems.

ID	age	income	student	credit rating	buy computer
1	<30	high	no	fair	no
2	<30	high	no	excellent	no
3	30-40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	30-40	low	yes	excellent	yes
8	<30	medium	no	fair	no
9	<30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<30	medium	yes	excellent	yes
12	30-40	medium	no	excellent	yes
13	30-40	high	yes	fair	yes
14	>40	medium	no	excellent	no

Table 2: Instances for naive Bayes

- (a) Computer the prior probability.
- (b) Computer the conditional probability of "age".
- (c) Given the new samples: $x = (\text{age} < 30, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit rating} = \text{fair})$.
Based on Naive Bayesian calculations, it determines whether to purchase a computer.

Ans:

(a)

(b)

(c)

3. Perform clustering analysis on the following two-dimensional data using the k-medoids algorithm. Assuming $k=2$, initialize the cluster centers using data points G and H. Calculate the similarity between any two points using Manhattan distance.

ID	A	B	C	D	E	F	G	H
feature 1	1.2	2.1	1.9	2.3	3.1	4.2	1.2	3.3
feature 2	1.1	2.4	3.1	4.2	4	4	1	2.2

Table 3: Instances for the k-medoids

- (a) Provide the clustering results and key computational steps.

Ans:

(a)